

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
7	(a)	(i)	Increase pd until microammeter reads zero. Or decrease pd until microammeter just starts to read a current (1) Reading of voltmeter <u>gives the energy value needed</u> or equivalent statement (1) Accept $E_k = eV_{\text{stop}}$	2			2		2
		(ii)	$\phi = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{405 \times 10^{-9}} \text{ [J] (1) } - 0.480 \times 1.60 \times 10^{-19} \text{ [J] (1)}$ $\text{or } \phi = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{405 \times 10^{-9} \times 1.60 \times 10^{-19}} \text{ [eV] (1) } - 0.480 \text{ [eV] (1)}$ [3.07] even if not transposed 4.1 × 10⁻¹⁹ [J] or 2.6 [eV] (1)		3		3	3	3
		(iii)	For 650 nm, photon energy = 3.1 × 10⁻¹⁹ J or 1.9 eV (1) Photon energy not enough to knock out electron or photon energy < ϕ (1) More photons [per second] won't help or increasing intensity doesn't increase photon energy (1) Accept only gives more photons at the same energy or doesn't change frequency Alternative Calculation of $f_0 = 6.2 \times 10^{14}$ [Hz] ecf or calculation of $\lambda_0 = 4.8 \times 10^{-7}$ [m] ecf (1) 4.6 × 10¹⁴ [Hz] < 6.2 × 10¹⁴ [Hz] or 650 nm is > 480 nm so no emission occurs (1) More photons [per second] won't help or increasing intensity doesn't increase photon energy (1) Accept only gives more photons at the same energy or doesn't change frequency - Subtract 1 mark if no useful reference made to photons			3	3		3